PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	Make **2-3 notes** of specific business requirements that will be analyzed. The transactions are properly handled and message sellers with questions. And also allows users to rate how the seller has performed for a good service. Behind the transactions, they ensure both user and seller have good interaction, and have proper back end processing. There aren't detailed regulations that need to be considered here.
II. Define the technical scope	List of technologies used by the application: • *Application programming interface (API)* • *Public key infrastructure (PKI)* • *SHA-256* • *SQL* I would prefer PKI Public key infrastructure because of the secure encryption framework that has both symmetric and asymmetric encryption algorithms, and it is considered strong in this situation due to hackers won't be able to get or obtain the key to decrypt sensitive information easily.
III. Decompose application	Sample data flow diagram
IV. Threat analysis	List **2 types of threats** in the PASTA worksheet that are risks to the information being handled by the application. • *What are the internal threats?* • *What are the external threats?* Based on external threats, there could be an injection using SQL language, and do something like login SQL injection to obtain credentials like username and password. For internal threats, insider workers like coworker might sell or

	disclose sensitive information after hating the company policy, or impacting his personal life, etc.
V. Vulnerability analysis	List **2 vulnerabilities** in the PASTA worksheet that could be exploited. Based on the scenario, there is a huge vulnerability of SQL injection that the login webpage could be altered by changing the website directory, or session tokens, certificates, etc. Weakness in this webpage like disclosing PII is critical and that makes the database vulnerable for outside attacks. Attackers can come from both inside and outside, misusing information like PII from inside and outside like flawed source code to inject malicious code.
VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	List **4 security controls** that you've learned about that can reduce risk. MFA multi factor authentication is a key way to control attackers to access anything. They verify if the user is authorized to enter or not, and check personal questions to match with the database. One could be separation of duties where you don't have unnecessary roles and privileges to anyone. Another one is a regular update or patch to change how people login and change source code, make sure you test all the code thoroughly. Adding firewalls in network traffic and make sure incoming and outgoing traffic networks are legit.